# INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

## SBT-DF204: Computer Forensics Case Studies

## Case Study 2: Reconstructing Suspected Drug Trafficking From Chrome Web History

*Individual Forensic Investigation Report*

| Field | Detail |
|---|---|
| **Student Name** | Mary-Joy Adewole |
| **Registration Number** | 2025/FWSD/11468 |
| **Institution** | International Cybersecurity and Digital Forensics Academy (ICDFA) |
| **Course Code** | SBT-DF204 |
| **Course Title** | Computer Forensics Case Studies |
| **Assessment** | Case Study 2: Individual forensic investigation |
| **Case Title** | Reconstructing Suspected Drug Trafficking From Chrome Web History |
| **Evidence File** | Google Chrome "History" SQLite database (E-Campus study material), 196,608 bytes |
| **Analysis Date** | 10 October 2026 (evidence copied 06:06:35 UTC); report dated 11 October 2026 |
| **Tools** | Kali Linux 2026.2 (VMware Workstation); sqlite3 3.46.1; sha256sum; md5sum; stat; file; date |
| **Time Basis** | All event times in this report are UTC, converted from Chrome WebKit timestamps. Any local-time remark is labelled as an assumption. |

## 1. Case Reference and Student Identification

This is the individual submission of Mary-Joy Adewole (registration number 2025/FWSD/11468) for SBT-DF204 Case Study 2. The scenario is a training exercise: a Google Chrome browsing-history database from a Windows profile is examined to reconstruct whether it shows the creation of an online classified advertisement, correspondence with a buyer, a claimed payment and an attempt to verify that payment, in connection with suspected drug trafficking.

The work was done on a read-only forensic copy of the supplied database, in a Kali Linux virtual machine, using SQLite queries only. In line with the case brief, nothing in the history was visited or contacted: the Craigslist posting, the Gmail account, the Imgur link, the blockchain addresses and transaction ID, Kraken and mempool.space appear in this report only as text recorded in the database. The original database is not included in the evidence package.

| Item | Detail |
|---|---|
| **Evidence item** | E01: Chrome "History" SQLite database (single file, no journal or WAL file supplied) |
| **Browser profile indicators** | Download path `C:\Users\FSCS_User\Desktop\proof_of_payment.png` (Windows account FSCS_User); Gmail page titles show the account unsub.fscs@gmail.com |
| **Activity window (UTC)** | 2022-04-19 13:54:18.232 to 15:02:56.453 (1 h 8 min 38.2 s), 74 visits, 57 URLs, 1 download |
| **Examiner** | Mary-Joy Adewole, 2025/FWSD/11468 |

## 2. Executive Conclusion

The History database is structurally sound (integrity_check = ok) and its 74 visits can be placed on a single UTC timeline. They show one continuous session on 19 April 2022 in which the same browser profile did the following, in order:

1. **Created a classified ad on Craigslist Baltimore** (13:54 to 13:58 UTC): signed in, stepped through the posting wizard, reached the confirmation page and opened the manage page and public listing for posting ID 7473121658, titled "cheaper than Rx supplements" in the health and beauty (by owner) category.
2. **Signed in to Gmail as unsub.fscs@gmail.com** (13:59:29 UTC) and later opened a conversation whose subject is "cheaper than Rx supplements - order" (14:04:12) and opened two compose windows in it.
3. **Opened a second conversation whose subject contains a 64-character transaction ID** (14:56:47), followed a link to Imgur and **saved an image as `proof_of_payment.png`** on the FSCS_User desktop (14:56:58 to 14:57:10).
4. **Looked up the same transaction ID, and one address, on a blockchain explorer** (14:59:06 to 14:59:21), **then signed in to Kraken** (new-device approval at 15:00:21), viewed Funding and Ledger pages and followed a Kraken link to the same transaction ID at mempool.space (15:01:31).
5. **Returned to the transaction-ID conversation and opened two compose windows** (15:02:32 to 15:02:44).

Read together, this is consistent with a seller advertising a product online, receiving an order, being sent a payment screenshot with a transaction ID, and checking that payment in a blockchain explorer and an exchange account. The History records navigation only. It does not record the advertisement text, the product offered, the email contents, whether any message was sent, or the amount and ownership of any payment, so it does not by itself prove that drugs were sold.

**Confidence.** High for what this browser profile did and when (database integrity, hashes and the UTC conversion were verified twice). Moderate for linking the payment evidence to the advertisement (the link rests on subject lines, the order of events and an exact transaction-ID match). Low for attributing the activity to a named person: the evidence identifies a browser profile, a Windows account name and an email account, not an individual. Section 5.7 lists the limitations and the follow-up needed.

## 3. Evidence Acquisition and Integrity

The supplied file was copied from the study-material download (held in the VMware shared folder) into a case folder, hashed immediately, made read-only, and only then was a working copy taken. All analysis used the working copy, opened with `sqlite3 -readonly`. The record below comes from the terminal output in Figures 1 to 4.

| Attribute | Value |
|---|---|
| **Original file name** | History (no extension; Chrome profile database) |
| **Source** | E-Campus study material for SBT-DF204 Case Study 2, downloaded to Windows and read from the VMware shared folder `/mnt/hgfs/kali_documents/History` |
| **Copied to** | `~/SBTDF204_CS2/original/History` using `cp -p` (preserves timestamps) |
| **File size** | 196,608 bytes (`ls -l`, `stat`) |

| Attribute | Value |
|---|---|
| **Acquisition date/time** | Saturday 10 October 2026, 06:06:35 UTC (`date -u`; the Kali VM clock is UTC-4, so `stat` shows 02:06:35 -0400) |
| **SHA-256** | b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 |
| **MD5** | 0c594e9de87348c089554953a67e2c74 |
| **File type (`file`)** | SQLite 3.x database, last written using SQLite version 3036000, file counter 33, database pages 41, cookie 0xa7, schema 4, UTF-8, version-valid-for 33 |
| **Original protection** | `chmod a-w original/History`; permissions afterwards `-r-xr-xr-x` |
| **Working copy** | `cp -p original/History working/History`; SHA-256 identical to the original ("HASHES MATCH", Figure 3) |
| **Integrity after analysis** | SHA-256 of `working/History` re-computed after all queries is unchanged (Figure 13); `working/` holds only `History`, so no `-journal` or `-wal` file was created |
| **Access method** | `sqlite3 -readonly` for every query; no write, VACUUM or PRAGMA that changes the file was run |

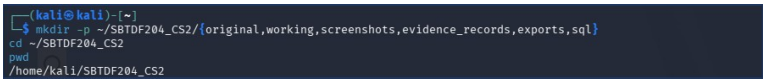



*Figure 1: Case folder created (original, working, screenshots, evidence_records, exports, sql). (E01)*

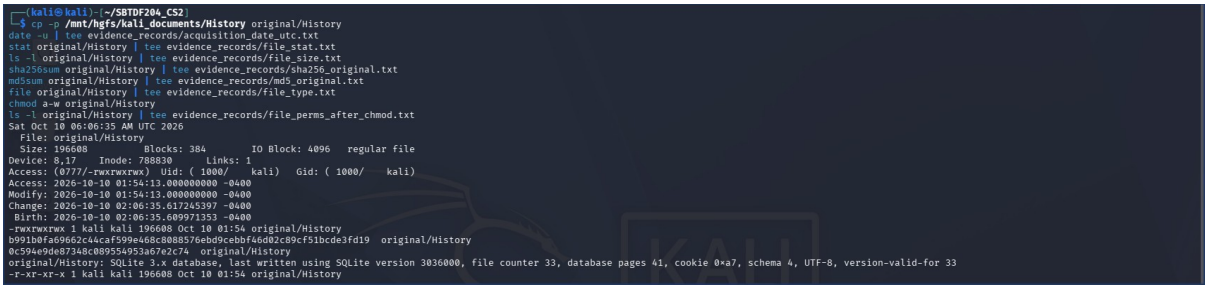



*Figure 2: Original History copied with cp -p, then date -u, stat, ls -l, sha256sum, md5sum, file and chmod a-w, each saved to evidence_records. (E01)*




*Figure 3: Working copy created with cp -p and its SHA-256 compared with the original ("HASHES MATCH"). (E02)*




*Figure 4: sqlite3 version used for all queries (3.46.1). (E02)*

## 4. Method

Each stage was checked before the next began, and every query was saved as a file in `sql/` with its output in `evidence_records/` (Appendix C).

- Preserve and verify the evidence (Section 3), then work only on the copy.
- Read the schema (`sqlite_master`, `PRAGMA table_info`) and row counts to see which tables hold navigation, download and search evidence, and check health with `PRAGMA integrity_check`, page size and page count.
- Convert Chrome WebKit timestamps (microseconds since 1601-01-01 UTC) to UTC inside the query, and cross-check one value independently with the shell `date` command.

- Join `visits` to `urls` so every visit shows visit ID, URL ID, time, `from_visit`, transition, duration, title and URL. Visit IDs and URL IDs are different numbers and both are cited.
- Decode the transition integer: `transition & 255` gives the core type, and the high bits give qualifiers such as chain start/end and client/server redirect. Chrome stores the value as a signed 32-bit integer, so values with the top bit set appear negative; the qualifiers are taken with the mask 4294967040 (0xFFFFFF00).
- Rebuild navigation paths with a recursive query on `from_visit`, and list `opener_visit` for links opened in a new tab or window.
- Read the `downloads` and `downloads_url_chains` tables, the `keyword_search_terms` table, and the `content_annotations` and `context_annotations` tables for supporting detail.
- Separate what the record shows (observed fact) from what it may mean (interpretation), and state the limit of each inference.

# 5. Findings

## 5.1 Database structure and fields (Question 2)

The database has 16 tables (Figure 5). The four the brief names, and the extra tables used as supporting evidence, are summarised below. Row counts are from Figure 7.

| Table | Rows | Fields used in this analysis | What it holds |
|---|---|---|---|
| urls | 57 | id, url, title, visit_count, typed_count, last_visit_time, hidden | One row per distinct URL. The title is stored here, once per URL, so it reflects the most recent load of that URL and not necessarily each visit. |
| visits | 74 | id, url (foreign key to urls.id), visit_time, from_visit, transition, visit_duration, opener_visit, segment_id, incremented_omnibox_typed_score | One row per page visit, with the navigation link (from_visit), how the page was reached (transition) and how long it was shown. |
| downloads | 1 | id, guid, target_path, current_path, start_time, end_time, last_access_time, received_bytes, total_bytes, state, danger_type, interrupt_reason, opened, mime_type, referrer, tab_url, hash | One row per download, including the local save path. |
| downloads_url_chains | 1 | id, chain_index, url | The URL (and any redirects) the download was fetched from, linked by download id. |
| keyword_search_terms | 4 | url_id, term, normalized_term | Search terms typed into the omnibox, linked to the results-page URL. |
| content_annotations / context_annotations | 47 / 45 | visit_id, entities, related_searches, search_terms; page_end_reason, total_foreground_duration, duration_since_last_visit | Chrome's page-content and usage annotations per visit (supporting only). |
| meta | 4 | key, value | Database version (53) and compatibility (16). |
| visit_source | 0 | id, source | Empty (see note below). |

`visits.url` is a foreign key to `urls.id`, and `downloads_url_chains.id` is a foreign key to `downloads.id`. The empty `visit_source` table is consistent with every visit being local browsing: Chrome writes rows there only for visits that were synced or imported, so I treat its emptiness as supporting, not conclusive. The domain summary (Figure 11) shows where the 74 visits went: mail.google.com 16, www.google.com 12, post.craigslist.org 10, www.kraken.com 10,

accounts.google.com 8, www.blockchain.com 6, and a few single-visit hosts including imgur.com and mempool.space.

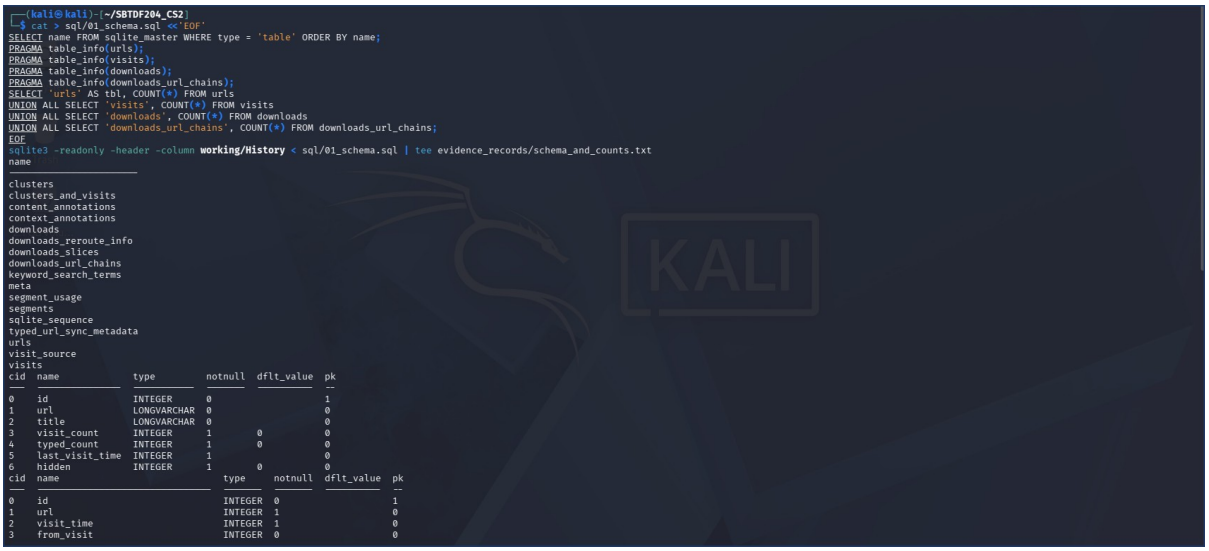



*Figure 5: Table list and the schema of urls and visits (start). (E03)*




*Figure 6: Rest of the visits schema, then the downloads and downloads_url_chains schemas and the four key row counts (urls 57, visits 74, downloads 1, downloads_url_chains 1). (E03)*




*Figure 7: Row count of every table in the database. (E04)*

## Database health

`PRAGMA integrity_check` returns **ok**; page size is 4096; the header page count is 41 and the free-list count is 0; encoding is UTF-8; `meta` shows version 53 and last_compatible_version 16 (Figure 8). The file is 196,608 bytes, which is 48 pages of 4,096 bytes, so there are 7 pages (28,672 bytes) after the 41 pages the header declares. I checked whether those bytes hold anything and found **0 non-zero bytes** (Figure 9), so they are zero padding and do not hide data.




*Figure 8: Page size, page count, free-list count, encoding, integrity_check and meta table. (E04)*

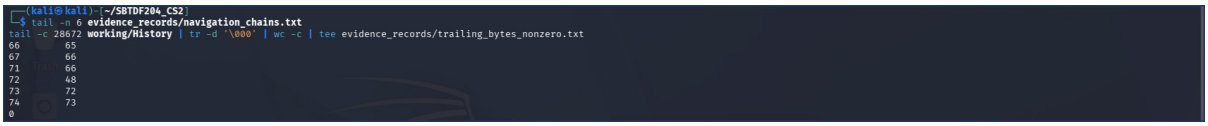



*Figure 9: The 28,672 bytes after the declared 41 pages contain 0 non-zero bytes (the final "0"). The lines above are the tail of the earlier navigation-chain output. (E05)*

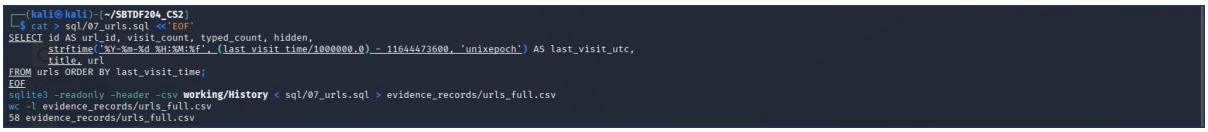



*Figure 10: Export of all 57 URLs with visit_count, typed_count, hidden and last-visit time to urls_full.csv (58 lines including the header). (E07)*

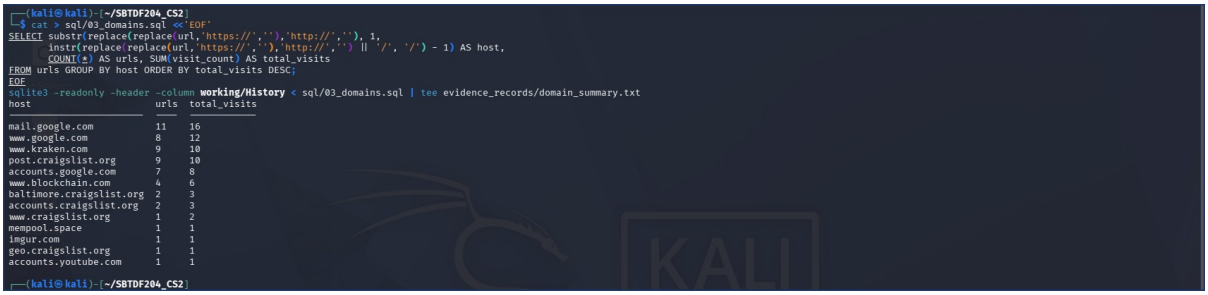



*Figure 11: Distinct URLs and total visits per host. (E23)*

## 5.2 Converting WebKit timestamps to UTC (Question 3)

Chrome stores `visit_time`, `start_time`, `end_time` and `last_visit_time` as the number of microseconds since 1601-01-01 00:00:00 UTC (the WebKit/Windows FILETIME epoch). The gap between that epoch and the Unix epoch (1970-01-01) is 11,644,473,600 seconds, so:

```
unix_seconds = webkit_microseconds / 1,000,000 - 11,644,473,600
strftime('%Y-%m-%d %H:%M:%f', (visit_time/1000000.0) - 11644473600, 'unixepoch')
```

**Worked example (visit 1).** Raw `visit_time` = 13294850058231547. Divided by 1,000,000 this is 13,294,850,058.231547 s. Subtracting 11,644,473,600 s gives 1,650,376,458.231547 Unix seconds, which is **2022-04-19 13:54:18.232 UTC**. The SQL conversion in the timeline export gives the same value, and an independent shell calculation, `date -u -d "@$((RAW/1000000 - 11644473600))"`, gives 2022-04-19 13:54:18 UTC (Figure 13). Because Chrome writes UTC, no time-zone adjustment is applied; the Kali VM's own clock (UTC-4) affects only the acquisition timestamps in Section 3, which I recorded with `date -u`.

The `downloads` table uses the same epoch. A zero value converts to 1601-01-01 00:00:00, which means "not set" and not a real date; `last_access_time` of download 1 is zero for that reason (Section 5.5).

**Local-time remark (assumption only).** The Craigslist site is the Baltimore one and Google's related-search suggestions mention Columbia, MD, so the machine may have been on US Eastern time (UTC-4 in April). If so, 13:54:18 UTC would be 09:54:18 local. Nothing in the database confirms the machine's time zone, so every conclusion in this report uses UTC.

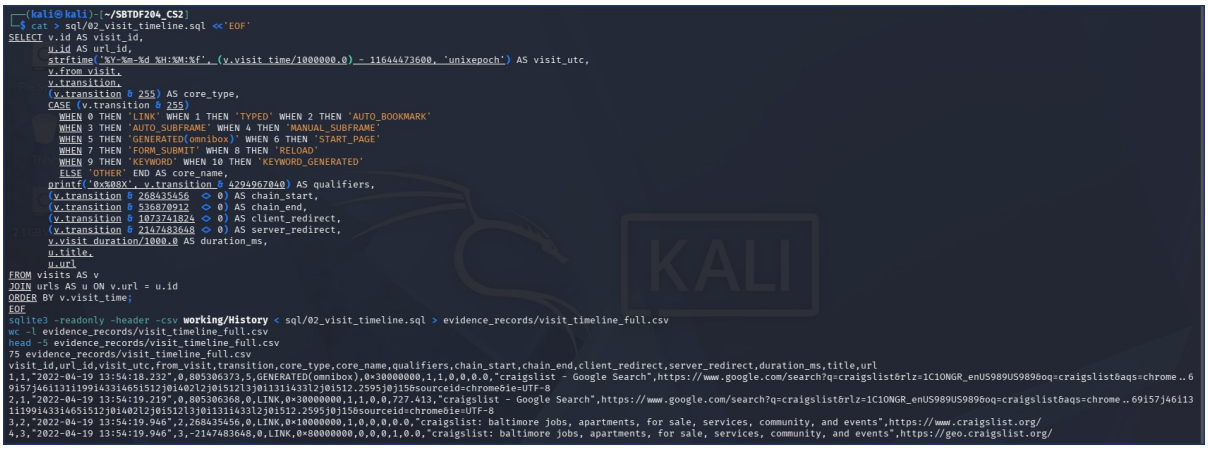



*Figure 12: SQL used to build the reconstructed timeline (02_visit_timeline.sql), with the UTC conversion, transition decoding and redirect flags; output saved to visit_timeline_full.csv (75 lines = header + 74 visits). (E06, E07)*




*Figure 13: Independent cross-check: raw visit_time 13294850058231547 converted with the shell date command gives 2022-04-19 13:54:18 UTC; working-copy SHA-256 re-checked after all queries (unchanged) and working/ directory listing. (E02, E06)*

## 5.3 Transitions, from_visit and navigation chains (Question 4)

Each row in `visits` carries `from_visit` (the visit it was reached from, 0 if none) and `transition` (how it was reached). Of the 74 visits, the core types are:

| Core type (transition & 255) | Count | Where it appears |
|---|---|---|
| 0 LINK | 63 | Clicked links, redirects and page loads that follow another page |
| 7 FORM_SUBMIT | 7 | Craigslist login submit (V8) and the posting-wizard steps (V14 to V19) |
| 5 GENERATED (omnibox search) | 4 | The four Google searches: craigslist (V1), gmail (V22), blockchain explorer (V51), kraken (V59) |
| 1 TYPED, 2 AUTO_BOOKMARK, 8 RELOAD | 0 | Not present: no typed URLs, bookmark opens or reloads |

The qualifier bits show how the visits were chained. Counted across the 74 visits, the qualifier values are 0x30000000 (chain start and end, a standalone navigation) 29, 0x80000000 (server redirect in the middle of a chain) 13, 0x10000000 (chain start) 12, 0xA0000000 (chain end after a server redirect) 9, 0x40000000 (client redirect) 5, 0x32000000 (chain start and end, typed in the address bar) 3, and 0x60000000 (chain end after a client redirect) 3. The three 0x32000000 visits are the omnibox searches V22, V51 and V59; V1 lacks the address-bar bit.

A recursive query on `from_visit` (Figure 14) groups the visits into 21 chains with no orphan links (the orphan check returned no rows). `opener_visit` (Figure 15) lists 28 visits opened from another visit, which is how links that open in a new tab or window are tied back to the page that opened them. The paths that matter for the case are:

| Path | Visits | Reading |
|---|---|---|
| Craigslist posting wizard | V2 > V3 > ... > V20 (one 19-visit chain) | Search result, Craigslist home, login, posting steps, confirmation and manage page are linked end to end by from_visit. The public listing V21 has from_visit 0 but opener_visit 20: it was opened from the manage page. |
| Gmail sign-in | V23 > V24 > V25 to V30; V36 > V37 to V42 | The first Gmail attempt is a seven-visit redirect chain that ends on the public Gmail "about" page. After the password step, a second chain (CheckCookie, SetOSID, SetSID, mail) ends on the inbox. |
| Gmail to Imgur | V48 (opener) > V49 > V50 | V49 is a Google link-redirect URL with `source=gmail`, opened from the txid conversation V48, then a server redirect to the Imgur page V50. |
| Blockchain explorer | V54 > V55 > V56; V54 > V57; V54 > V58 | From the explorer home page V54 the transaction search V55 (which redirects to the transaction page V56), the transaction page V57 and the address page V58 all have from_visit 54. |
| Kraken | V63 > V64, V65, V66, V71; V66 (opener) > V67 > V68 > V69 > V70 | Trade, Instant, Funding and Ledger pages follow V63. V67 to V70 is a four-step redirect chain, opened from the Funding page V66, ending on mempool.space for the same transaction ID. |
| Return to Gmail | V48 > V72 > V73 > V74 | V72 has from_visit 48 although it occurred about 6 minutes after V48, so the txid conversation tab kept running while the exchange pages were open. |

**Limit.** `from_visit` and `opener_visit` follow Chrome's navigation graph, not the user's tabs: History does not store a tab ID, so I cannot say how many tabs were open. Redirect steps and single-page-application URL changes (Gmail, Kraken) create visits that the user did not choose separately.




*Figure 14: Recursive query that rebuilds navigation chains from from_visit, the orphan check (no rows returned), and the start of the opener_visit list. (E20)*

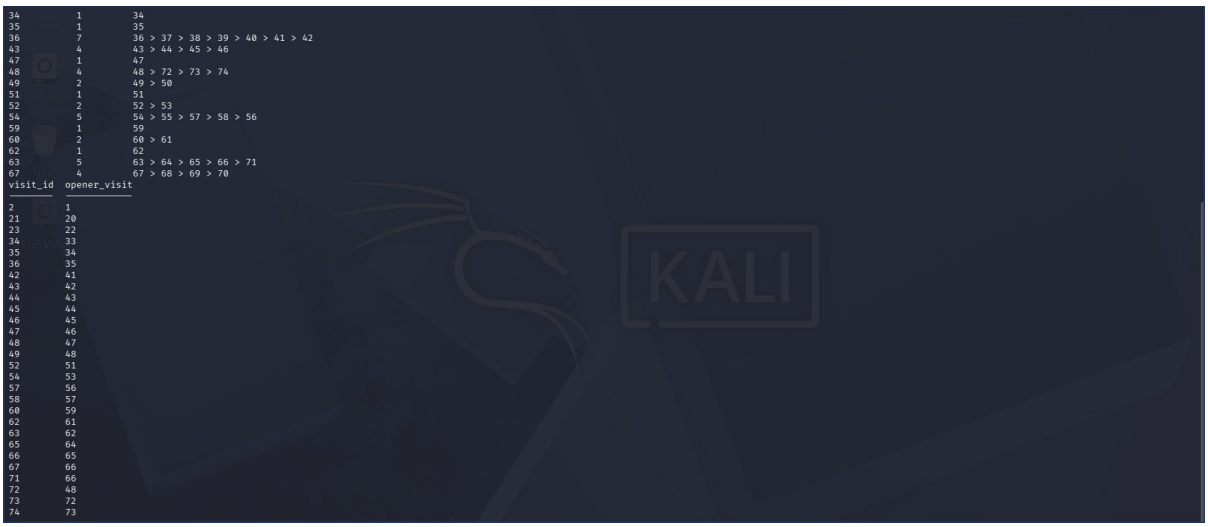



*Figure 15: Remaining chains and the complete opener_visit list (28 rows). (E20)*

## 5.4 Posting evidence (Question 5)

**Observed.** After a Google search for "craigslist" (V1, 13:54:18) the browser reached craigslist.org, was redirected through geo.craigslist.org to **baltimore.craigslist.org** (V3 to V5, all at 13:54:19.946), and requested the account page `accounts.craigslist.org/login/home`, which redirected to the login form (V6, V7). A form submit to `/login/home` follows at **13:54:39.868 (V8)**. The posting wizard then runs on `post.craigslist.org` under a single session key (`/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5`), each step reached by FORM_SUBMIT:

| Visit / URL | UTC | Wizard step (`?s=` value) | Time on page |
|---|---|---|---|
| V11 / U7 | 13:55:05.972 | post.craigslist.org/c/bal, "choose type" | (redirect) |
| V12 to V13 / U8, U9 | 13:55:05.972 | Server redirect assigns the session key; `?s=type` | 4.3 s |
| V14 / U10 | 13:55:10.245 | `?s=cat`, "choose category" | 5.6 s |
| V15 / U11 | 13:55:15.844 | `?s=edit`, "create posting" | 177.7 s (2 min 57.7 s) |
| V16 / U12 | 13:58:13.519 | `?s=geoverify`, "add map" | 4.4 s |
| V17 / U13 | 13:58:17.908 | `?s=editimage`, "create posting" | 1.6 s |
| V18 / U14 | 13:58:19.558 | `?s=preview`, "create posting" | 3.2 s |
| V19 / U8 | 13:58:22.757 | "posting confirmation" (same URL as V12) | 5.0 s |
| V20 / U15 | 13:58:27.747 | `post.craigslist.org/manage/7473121658`, "manage posting" | 5.6 s |
| V21 / U16 | 13:58:31.093 | `baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-supplements/7473121658.html` | (opened from V20) |

The same number, **7473121658**, is in the manage URL and in the public listing URL, which ties the two together. The public page title is "cheaper than Rx supplements - health and beauty - by owner -..." as stored in the database, and the path segment `/hab/` is Craigslist's health and beauty section. The listing was opened 3.3 s after the manage page, with the manage page as its opener.

**Interpretation.** The sequence of login, wizard steps, confirmation, manage page and public listing is the normal path for publishing a Craigslist ad, so this browser profile very probably created and published posting 7473121658 in the Baltimore health and beauty section. The 177.7 s on the "create posting" step (Chrome's foreground-time annotation for V15 is 178.0 s) fits

the time needed to write a title and description, and the next three steps (map, image, preview) were completed in under 10 s each.

**Limit.** History holds URLs and titles, not page content. It does not show what the ad text said, which image was uploaded, what was being sold, or whether the login at V8 succeeded (the login URL's later title is "craigslist account", which is consistent with success but is a per-URL title). The title's wording ("cheaper than Rx supplements") suggests a medication-substitute product, but that is an inference from a title, not proof of an illegal sale.

## 5.5 Communications, payment and download evidence (Question 6)

### Gmail sign-in (V22 to V42)

**Observed.** Seven seconds after opening the listing (V21, 13:58:31), a Google search for "gmail" was made (V22, 13:58:38.596). The first attempt to open Gmail (V24 to V30, all stamped 13:58:40.385) is a seven-visit redirect chain through `mail.google.com`, `accounts.google.com/ServiceLogin` and the Gmail help pages, ending on `www.google.com/gmail/about/`. The browser then went through the account chooser (V31, 13:58:43.290) to the sign-in identifier page (V35, 13:58:44.178) and, **42.2 s later, to the password challenge** (V36, 13:59:26.399). V36 was shown for 2,967.5 ms, which ends exactly at V37 (13:59:29.366), the `CheckCookie` page, followed by `SetOSID`, `SetSID` on accounts.youtube.com and the inbox (V42, 13:59:31.970).

**Interpretation.** The first chain ended on the public "about" page, so the browser was not signed in at 13:58:40. The identifier-to-password gap and the immediate CheckCookie redirect show a sign-in that completed at about 13:59:29. The titles of the inbox URLs read "Inbox (12) - unsub.fscs@gmail.com - Gmail", which identifies the account. **Limit:** titles are stored once per URL, so earlier visits to the same URLs (for example V25 and V26, which belong to the failed first attempt) display that later title.

### Order correspondence (V43 to V47)

**Observed.** At 14:04:12.566 (V43), 4 min 40.6 s after the inbox loaded, a conversation titled "cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail" was opened from the inbox (URL 34). Two URLs with a `?compose=` parameter follow at 14:04:35.250 (V44) and 14:04:46.777 (V45), each with a different compose token, then the conversation again (V46, 14:04:52.145) and the inbox (V47, 14:05:59.142). The History then has **no entry for 50 min 48.7 s**, until 14:56:47.857.

**Interpretation.** The subject repeats the advertisement title with the word "order", so a conversation about an order for the advertised product existed in this mailbox. The two compose URLs mean a compose or reply window was opened twice. **Limit:** History shows the subject only, not who wrote, what it said or whether a message was sent. The 50-minute gap may be idle time, another application, another browser or an incognito window.

### Transaction-ID conversation, Imgur link and download (V48 to V50, download 1)

**Observed.** At 14:56:47.857 (V48) a different Gmail conversation was opened (URL 37, a different thread identifier from V43). Its title is "cheaper than Rx supplements -- txid: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - unsub.fscs@gmail.com - Gmail". Five seconds later (V49, 14:56:52.464) a Google redirect URL `google.com/url?q=https://imgur.com/dTgrkP7&source=gmail` was opened, with V48 as its opener, and a server redirect took the browser to `imgur.com/dTgrkP7` (V50), where it stayed 18.9 s. The `downloads` table (Figure 19) records download 1 starting at **14:56:58.494**, 6.0 s after V50, and ending at **14:57:10.261** (11.8 s):

| Field | Value |
|---|---|
| download id / guid | 1 / 104c5d2b-4cba-40f3-9e11-02b782e54bfa |
| Fetched from (downloads_url_chains) | https://i.imgur.com/dTgrkP7.png (chain_index 0, no redirects) |
| Saved as (target_path = current_path) | C:\Users\FSCS_User\Desktop\proof_of_payment.png |
| Size | received_bytes 33,844 = total_bytes 33,844; state 1 (complete); interrupt_reason 0; danger_type 0 |
| MIME type | image/png (original_mime_type image/png) |
| referrer / tab_url | https://imgur.com/ / https://imgur.com/dTgrkP7 (the same page as V50) |
| opened / last_access_time | 0 / 0 (converts to 1601-01-01, meaning "not set") |
| hash | empty (Chrome did not store a file hash) |

**Interpretation.** The page chain Gmail txid conversation, Imgur link, Imgur page, download shows that the image was reached from a link inside the email and saved to the FSCS_User desktop under the name `proof_of_payment.png`. The saved name differs from the URL's file name (`dTgrkP7.png`), so it was set when the file was saved (a Save-as name or a server header); History does not say which. Its filename and the subject's "txid" are consistent with a claimed payment. **Limit:** the image itself was not available, so I cannot say what it shows, and `opened = 0` only means Chrome did not record opening it from Chrome.

### Blockchain explorer lookup (V51 to V58)

**Observed.** 1 min 49.3 s after the download finished, a Google search for "blockchain explorer" was made (V51, 14:58:59.580) and the browser reached `blockchain.com/explorer` (V53 and V54). At 14:59:06.190 the browser requested `blockchain.com/search?search=517b2156...afdc5` (V55), which redirected to `blockchain.com/btc/tx/517b2156...afdc5` (V56, V57), and at 14:59:21.433 it opened `blockchain.com/btc/address/38RcsURWCDmCbYocmsZCnGz1FtD7x477mt` (V58, title "Address: 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt | Blockchain Explorer"). The 64-character value searched is **identical** to the one in the title of V48; the transaction page was reached 2 min 18.3 s after V48.

**Interpretation.** The transaction ID from the email subject was copied into a blockchain explorer and the transaction, and one address, were looked up. **Limit:** I did not visit these pages, so I cannot say whether the transaction exists, which coins it moved, or whether the address belongs to the user, the buyer or anyone else. The address was opened from the explorer but History does not show why.

### Exchange account access (V59 to V71)

**Observed.** A Google search for "kraken" (V59, 14:59:38.025) led to `kraken.com/en-us/sign-in` (V61, V62, 14:59:47). At **15:00:21.272** the browser loaded `kraken.com/en-us/device-approval`, titled "Approve new device", and stayed on it 104.4 s. The Trade (V64, 15:01:11.671), Instant (V65) and Funding (V66, 15:01:21.602) pages followed. The Funding page title is "41,543.80 USD - Kraken - Funding - Overview" and the Ledger page (V71, 15:02:05.697) is "41,563.70 USD - Kraken - History - Ledger". At 15:01:31.926 a Kraken redirect URL carrying `mempool.space/tx/517b2156...afdc5` was opened from the Funding page (V67), passed through two further Kraken redirect steps (V68, V69) and ended on `mempool.space/tx/517b2156...afdc5` at 15:01:33.808 (V70).

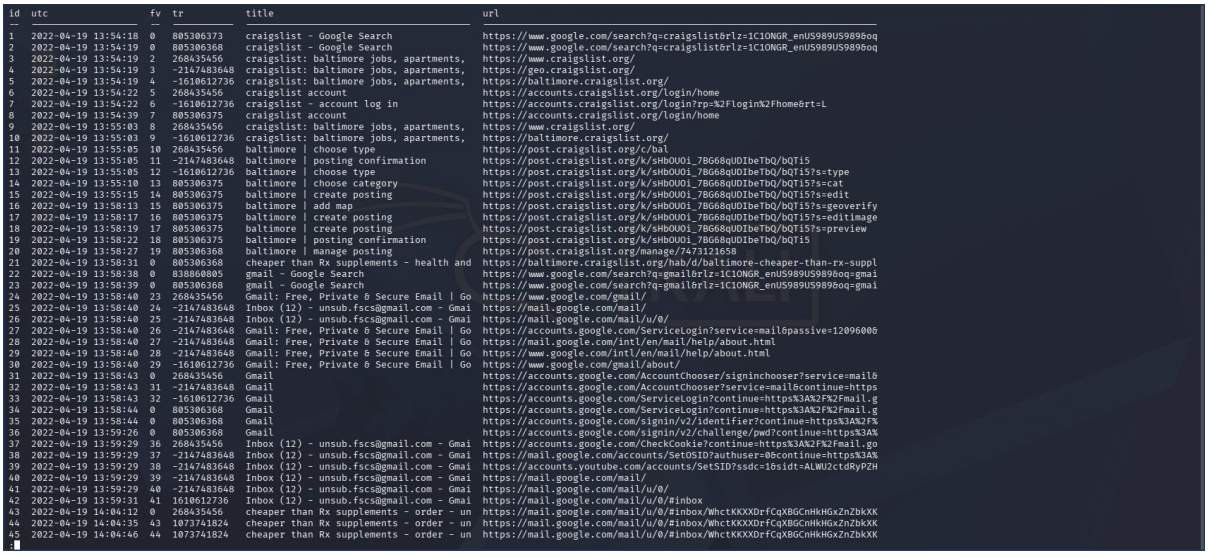
**Interpretation.** A Kraken account was accessed from this browser. The device-approval page means Kraken did not recognise the browser at that time, and the account-balance text in the page titles shows that the session was signed in. The outbound link from the Funding page to the same transaction ID suggests the transaction was visible from within the account (for example as a deposit), but History does not show that. **Limit:** titles are per URL and the USD figures are the last values stored for those URLs, so they are not balances at the time of each visit; no Gmail visit appears between V63 and V72, so how the new device was approved is not recorded; and the amounts and account holder are not in the database.

### Return to the txid conversation (V72 to V74)

**Observed.** At 15:02:32.450 (V72) the txid conversation was reached again with a `?compose=` token, 60.5 s after the mempool page, then a second compose URL at 15:02:44.171 (V73) and the conversation itself at 15:02:56.453 (V74). The compose URLs of V72 and V73 are for the txid thread but carry the title "cheaper than Rx supplements - order - ...", while V74 carries the "txid" title. **Interpretation:** a compose window was opened twice in the conversation after the payment checks. **Limit:** this is not evidence that a message was sent; and the title mismatch is an example of why titles alone should not be used to date content.

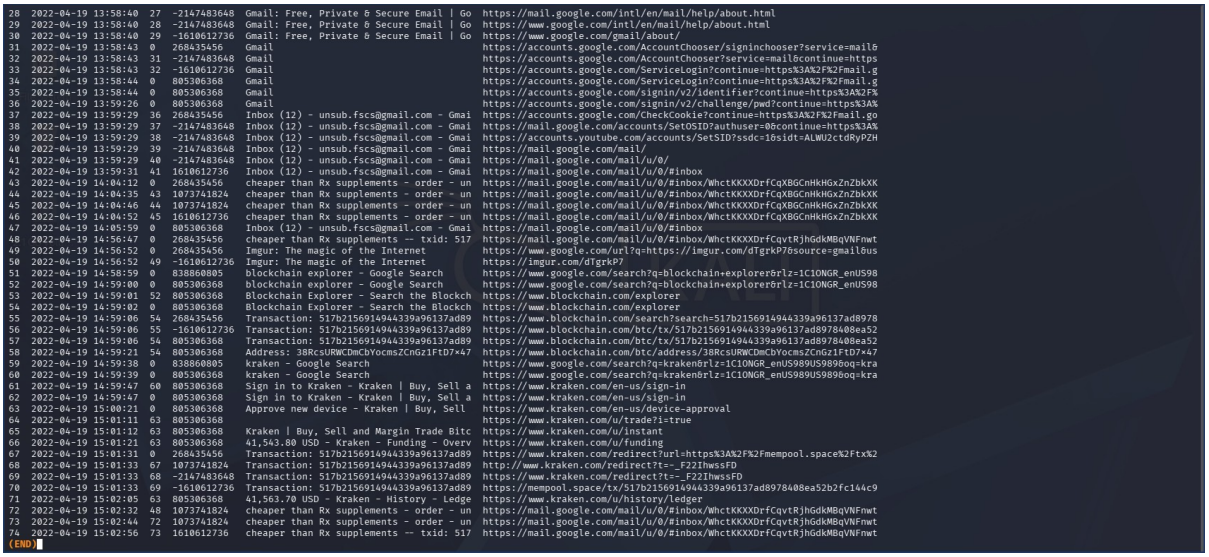
### Search terms and annotations

The four omnibox searches (Figure 18) are: craigslist (URL 1), gmail (URL 17), blockchain explorer (URL 40), kraken (URL 45). All are Google searches and each is followed within seconds by the service it names. The annotation tables (Figure 20) add supporting detail: Google's `related_searches` for V2 is "craigslist near columbia, md" (a weak location hint), and `context_annotations` records foreground time of 178.0 s for V15, 46.3 s for V33 (Gmail inbox) and 84.8 s for V61 (Kraken sign-in). Annotation values of -1000000 mean "not recorded".



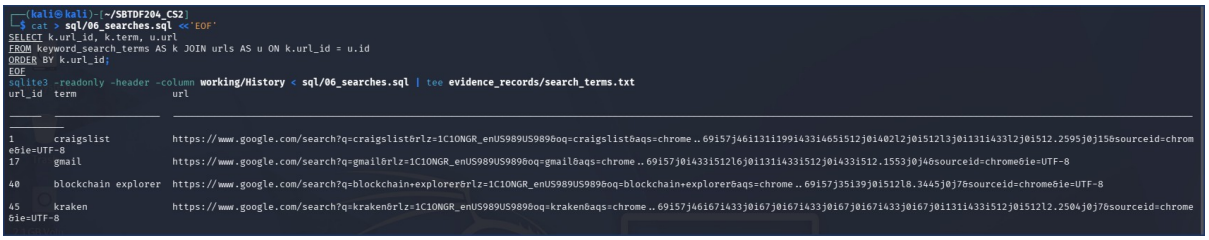
| id | utc | fv | tr | title | url |
|---|---|---|---|---|---|
| 1 | 2022-04-19 13:54:18 | 0 | 805306373 | craigslist - Google Search | https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq |
| 2 | 2022-04-19 13:54:19 | 0 | 805306368 | craigslist - Google Search | https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq |
| 3 | 2022-04-19 13:54:19 | 2 | 268435456 | craigslist: baltimore jobs, apartments, | https://www.craigslist.org/ |
| 4 | 2022-04-19 13:54:19 | 3 | -2147483648 | craigslist: baltimore jobs, apartments, | https://geo.craigslist.org/ |
| 5 | 2022-04-19 13:54:19 | 4 | -1610612736 | craigslist: baltimore jobs, apartments, | https://baltimore.craigslist.org/ |
| 6 | 2022-04-19 13:54:22 | 5 | 268435456 | craigslist account | https://accounts.craigslist.org/login/home |
| 7 | 2022-04-19 13:54:22 | 6 | -1610612736 | craigslist - account log in | https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L |
| 8 | 2022-04-19 13:54:39 | 7 | 805306375 | craigslist account | https://accounts.craigslist.org/login/home |
| 9 | 2022-04-19 13:55:03 | 8 | 268435456 | craigslist: baltimore jobs, apartments, | https://www.craigslist.org/ |
| 10 | 2022-04-19 13:55:03 | 9 | -1610612736 | craigslist: baltimore jobs, apartments, | https://baltimore.craigslist.org/ |
| 11 | 2022-04-19 13:55:05 | 10 | 268435456 | baltimore \| choose type | https://post.craigslist.org/c/bal |
| 12 | 2022-04-19 13:55:05 | 11 | -2147483648 | baltimore \| posting confirmation | https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5 |
| 13 | 2022-04-19 13:55:05 | 12 | -1610612736 | baltimore \| choose type | https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=type |
| 14 | 2022-04-19 13:55:10 | 13 | 805306375 | baltimore \| choose category | https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=cat |
| 15 | 2022-04-19 13:55:15 | 14 | 805306375 | baltimore \| create posting | https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=edit |
| 16 | 2022-04-19 13:58:13 | 15 | 805306375 | baltimore \| add map | https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=geoverify |
| 17 | 2022-04-19 13:58:17 | 16 | 805306375 | baltimore \| create posting | https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=editimage |
| 18 | 2022-04-19 13:58:19 | 17 | 805306375 | baltimore \| create posting | https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=preview |
| 19 | 2022-04-19 13:58:22 | 18 | 805306375 | baltimore \| posting confirmation | https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5 |
| 20 | 2022-04-19 13:58:27 | 19 | 805306368 | baltimore \| manage posting | https://post.craigslist.org/manage/7473121658 |
| 21 | 2022-04-19 13:58:31 | 0 | 805306368 | cheaper than Rx supplements - health and | https://baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-suppl |
| 22 | 2022-04-19 13:58:38 | 0 | 838860805 | gmail - Google Search | https://www.google.com/search?q=gmail&rlz=1C1ONGR_enUS989US989&oq=gmai |
| 23 | 2022-04-19 13:58:39 | 0 | 805306368 | gmail - Google Search | https://www.google.com/search?q=gmail&rlz=1C1ONGR_enUS989US989&oq=gmai |
| 24 | 2022-04-19 13:58:40 | 23 | 268435456 | Gmail: Free, Private & Secure Email \| Go | https://www.google.com/gmail/ |
| 25 | 2022-04-19 13:58:40 | 24 | -2147483648 | Inbox (12) - unsub.fscs@gmail.com - Gmai | https://mail.google.com/mail/ |
| 26 | 2022-04-19 13:58:40 | 25 | -2147483648 | Inbox (12) - unsub.fscs@gmail.com - Gmai | https://mail.google.com/mail/u/0/ |
| 27 | 2022-04-19 13:58:40 | 26 | -2147483648 | Gmail: Free, Private & Secure Email \| Go | https://accounts.google.com/ServiceLogin?service=mail&passive=1209600& |
| 28 | 2022-04-19 13:58:40 | 27 | -2147483648 | Gmail: Free, Private & Secure Email \| Go | https://mail.google.com/intl/en/mail/help/about.html |
| 29 | 2022-04-19 13:58:40 | 28 | -2147483648 | Gmail: Free, Private & Secure Email \| Go | https://www.google.com/intl/en/mail/help/about.html |
| 30 | 2022-04-19 13:58:40 | 29 | -1610612736 | Gmail: Free, Private & Secure Email \| Go | https://www.google.com/gmail/about/ |
| 31 | 2022-04-19 13:58:43 | 0 | 268435456 | Gmail | https://accounts.google.com/AccountChooser/signinchooser?service=mail& |
| 32 | 2022-04-19 13:58:43 | 31 | -2147483648 | Gmail | https://accounts.google.com/AccountChooser?service=mail&continue=https |
| 33 | 2022-04-19 13:58:43 | 32 | -1610612736 | Gmail | https://accounts.google.com/ServiceLogin?continue=https%3A%2F%2Fmail.g |
| 34 | 2022-04-19 13:58:44 | 0 | 805306368 | Gmail | https://accounts.google.com/ServiceLogin?continue=https%3A%2F%2Fmail.g |
| 35 | 2022-04-19 13:58:44 | 0 | 805306368 | Gmail | https://accounts.google.com/signin/v2/identifier?continue=https%3A%2F% |
| 36 | 2022-04-19 13:59:26 | 0 | 805306368 | Gmail | https://accounts.google.com/signin/v2/challenge/pwd?continue=https%3A% |
| 37 | 2022-04-19 13:59:29 | 36 | 268435456 | Inbox (12) - unsub.fscs@gmail.com - Gmai | https://accounts.google.com/CheckCookie?continue=https%3A%2F%2Fmail.go |
| 38 | 2022-04-19 13:59:29 | 37 | -2147483648 | Inbox (12) - unsub.fscs@gmail.com - Gmai | https://mail.google.com/accounts/SetOSID?authuser=0&continue=https%3A% |
| 39 | 2022-04-19 13:59:29 | 38 | -2147483648 | Inbox (12) - unsub.fscs@gmail.com - Gmai | https://accounts.youtube.com/accounts/SetSID?ssdc=1&sidt=ALWU2ctdRyPZH |
| 40 | 2022-04-19 13:59:29 | 39 | -2147483648 | Inbox (12) - unsub.fscs@gmail.com - Gmai | https://mail.google.com/mail/ |
| 41 | 2022-04-19 13:59:29 | 40 | -2147483648 | Inbox (12) - unsub.fscs@gmail.com - Gmai | https://mail.google.com/mail/u/0/ |
| 42 | 2022-04-19 13:59:31 | 41 | 1610612736 | Inbox (12) - unsub.fscs@gmail.com - Gmai | https://mail.google.com/mail/u/0/#inbox |
| 43 | 2022-04-19 14:04:12 | 0 | 268435456 | cheaper than Rx supplements - order - un | https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbkXK |
| 44 | 2022-04-19 14:04:35 | 43 | 1073741824 | cheaper than Rx supplements - order - un | https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbkXK |
| 45 | 2022-04-19 14:04:46 | 44 | 1073741824 | cheaper than Rx supplements - order - un | https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbkXK |

*Figure 16: Reconstructed visit timeline, visits 1 to 45 (visit ID, UTC time, from_visit, transition, title, URL). (E07)*



```
28  2022-04-19 13:58:40  27  -2147483648  Gmail: Free, Private & Secure Email | Go  https://mail.google.com/intl/en/mail/help/about.html
29  2022-04-19 13:58:40  28  -2147483648  Gmail: Free, Private & Secure Email | Go  https://www.google.com/intl/en/mail/help/about.html
30  2022-04-19 13:58:40  29  -1610612736  Gmail: Free, Private & Secure Email | Go  https://www.google.com/gmail/about/
31  2022-04-19 13:58:43  0   268435456    Gmail                                    https://accounts.google.com/AccountChooser/signinchooser?service=mail&
32  2022-04-19 13:58:43  31  -2147483648  Gmail                                    https://accounts.google.com/AccountChooser?service=mail&continue=https
33  2022-04-19 13:58:43  32  -1610612736  Gmail                                    https://accounts.google.com/ServiceLogin?continue=https%3A%2F%2Fmail.g
34  2022-04-19 13:58:44  0   805306368    Gmail                                    https://accounts.google.com/ServiceLogin?continue=https%3A%2F%2Fmail.g
35  2022-04-19 13:58:44  0   805306368    Gmail                                    https://accounts.google.com/signin/v2/identifier?continue=https%3A%2F%
36  2022-04-19 13:59:26  0   805306368    Gmail                                    https://accounts.google.com/signin/v2/challenge/pwd?continue=https%3A%
37  2022-04-19 13:59:29  36  268435456    Inbox (12) - unsub.fscs@gmail.com - Gmai  https://accounts.google.com/CheckCookie?continue=https%3A%2F%2Fmail.go
38  2022-04-19 13:59:29  37  -2147483648  Inbox (12) - unsub.fscs@gmail.com - Gmai  https://mail.google.com/accounts/SetOSID?authuser=0&continue=https%3A%
39  2022-04-19 13:59:29  38  -2147483648  Inbox (12) - unsub.fscs@gmail.com - Gmai  https://accounts.youtube.com/accounts/SetSID?ssdc=1&sidt=ALWU2ctdRyP2H
40  2022-04-19 13:59:29  39  -2147483648  Inbox (12) - unsub.fscs@gmail.com - Gmai  https://mail.google.com/mail/
41  2022-04-19 13:59:29  40  -2147483648  Inbox (12) - unsub.fscs@gmail.com - Gmai  https://mail.google.com/mail/u/0/
42  2022-04-19 13:59:31  41  1610612736   Inbox (12) - unsub.fscs@gmail.com - Gmai  https://mail.google.com/mail/u/0/#inbox
43  2022-04-19 14:04:12  0   268435456    cheaper than Rx supplements - order - un  https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbkKK
44  2022-04-19 14:04:35  43  1073741824   cheaper than Rx supplements - order - un  https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbkKK
45  2022-04-19 14:04:46  44  1073741824   cheaper than Rx supplements - order - un  https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbkKK
46  2022-04-19 14:04:52  45  1610612736   cheaper than Rx supplements - order - un  https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbkKK
47  2022-04-19 14:05:59  0   805306368    Inbox (12) - unsub.fscs@gmail.com - Gmai  https://mail.google.com/mail/u/0/#inbox
48  2022-04-19 14:56:47  0   268435456    cheaper than Rx supplements -- txid: 517  https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtRjhGdkMBqVNFnwt
49  2022-04-19 14:56:52  0   268435456    Imgur: The magic of the Internet          https://www.google.com/url?q=https://imgur.com/dTgrkP7&source=gmail&us
50  2022-04-19 14:56:52  49  -1610612736  Imgur: The magic of the Internet          https://imgur.com/dTgrkP7
51  2022-04-19 14:58:59  0   838860805    blockchain explorer - Google Search       https://www.google.com/search?q=blockchain+explorer&rlz=1C1ONGR_enUS98
52  2022-04-19 14:59:00  0   805306368    blockchain explorer - Google Search       https://www.google.com/search?q=blockchain+explorer&rlz=1C1ONGR_enUS98
53  2022-04-19 14:59:01  52  805306368    Blockchain Explorer - Search the Blockch  https://www.blockchain.com/explorer
54  2022-04-19 14:59:02  0   805306368    Blockchain Explorer - Search the Blockch  https://www.blockchain.com/explorer
55  2022-04-19 14:59:06  54  268435456    Transaction: 517b2156914944339a96137ad89  https://www.blockchain.com/search?search=517b2156914944339a96137ad8978
56  2022-04-19 14:59:06  55  -1610612736  Transaction: 517b2156914944339a96137ad89  https://www.blockchain.com/btc/tx/517b2156914944339a96137ad8978408ea52
57  2022-04-19 14:59:06  54  805306368    Transaction: 517b2156914944339a96137ad89  https://www.blockchain.com/btc/tx/517b2156914944339a96137ad8978408ea52
58  2022-04-19 14:59:21  54  805306368    Address: 38RcsURWCDmCbYocmsZCnGz1FtD7×47  https://www.blockchain.com/btc/address/38RcsURWCDmCbYocmsZCnGz1FtD7×47
59  2022-04-19 14:59:38  0   838860805    kraken - Google Search                    https://www.google.com/search?q=kraken6rlz=1C1ONGR_enUS989US989&oq=kra
60  2022-04-19 14:59:39  0   805306368    kraken - Google Search                    https://www.google.com/search?q=kraken6rlz=1C1ONGR_enUS989US989&oq=kra
61  2022-04-19 14:59:47  60  805306368    Sign in to Kraken - Kraken | Buy, Sell a  https://www.kraken.com/en-us/sign-in
62  2022-04-19 14:59:47  0   805306368    Sign in to Kraken - Kraken | Buy, Sell a  https://www.kraken.com/en-us/sign-in
63  2022-04-19 15:00:21  0   805306368    Approve new device - Kraken | Buy, Sell   https://www.kraken.com/en-us/device-approval
64  2022-04-19 15:01:11  63  805306368                                             https://www.kraken.com/u/trade?i=true
65  2022-04-19 15:01:12  63  805306368    Kraken | Buy, Sell and Margin Trade Bitc  https://www.kraken.com/u/instant
66  2022-04-19 15:01:21  63  805306368    41,543.80 USD - Kraken - Funding - Overv  https://www.kraken.com/u/funding
67  2022-04-19 15:01:31  0   268435456    Transaction: 517b2156914944339a96137ad89  https://www.kraken.com/redirect?url=https%3A%2F%2Fmempool.space%2Ftx%2
68  2022-04-19 15:01:33  67  1073741824   Transaction: 517b2156914944339a96137ad89  http://www.kraken.com/redirect?t=-_F22IhwssFD
69  2022-04-19 15:01:33  68  -2147483648  Transaction: 517b2156914944339a96137ad89  https://www.kraken.com/redirect?t=-_F22IhwssFD
70  2022-04-19 15:01:33  69  -1610612736  Transaction: 517b2156914944339a96137ad89  https://mempool.space/tx/517b2156914944339a96137ad8978408ea52b2fc144c9
71  2022-04-19 15:02:05  63  805306368    41,563.70 USD - Kraken - History - Ledge  https://www.kraken.com/u/history/ledger
72  2022-04-19 15:02:32  48  1073741824   cheaper than Rx supplements - order - un  https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtRjhGdkMBqVNFnwt
73  2022-04-19 15:02:44  72  1073741824   cheaper than Rx supplements - order - un  https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtRjhGdkMBqVNFnwt
74  2022-04-19 15:02:56  73  1610612736   cheaper than Rx supplements -- txid: 517  https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtRjhGdkMBqVNFnwt
(END)
```

*Figure 17: Reconstructed visit timeline, visits 28 to 74. (E07)*



```
┌──(kali㉿kali)-[~/SBTDF204_CS2]
└─$ cat > sql/06_searches.sql <<'EOF'
SELECT k.url_id, k.term, u.url
FROM keyword_search_terms AS k JOIN urls AS u ON k.url_id = u.id
ORDER BY k.url_id;
EOF
sqlite3 -readonly -header -column working/History < sql/06_searches.sql | tee evidence_records/search_terms.txt
url_id  term                url
──────  ──────────────────  ──────────────────
```

## 5.6 Reconstructed timeline (Question 7)

The table summarises the ten stages. The visit-by-visit version, with observed fact separated from interpretation and limit, is in Appendix A, and every visit is listed in Appendix E. All times are UTC on 2022-04-19.

| Stage | UTC window | Visits | What the record shows | Reading |
|---|---|---|---|---|
| S1 Search and enter Craigslist | 13:54:18 to 13:54:20 | V1 to V5 | Google search "craigslist", redirect to baltimore.craigslist.org | Start of session; Baltimore site |
| S2 Craigslist login | 13:54:22 to 13:54:40 | V6 to V8 | Login page, form submit | Account login attempted |
| S3 Create posting | 13:55:03 to 13:58:23 | V9 to V19 | Wizard steps type, cat, edit (177.7 s), map, image, preview, confirmation | Ad created in this profile |
| S4 Check listing | 13:58:27 to 13:58:31 | V20, V21 | Manage page and public listing, ID 7473121658 | Ad live, same ID |
| S5 Gmail sign-in | 13:58:38 to 13:59:32 | V22 to V42 | Search, failed redirect, identifier, password, inbox (unsub.fscs@gmail.com) | Mailbox accessed |
| S6 Order thread | 14:04:12 to 14:05:59 | V43 to V47 | Conversation "... - order", two compose URLs | Correspondence about an order |
| (gap) | 14:05:59 to 14:56:47 | none | No history for 50 min 48.7 s | Unexplained (limit) |
| S7 Payment proof | 14:56:47 to 14:57:10 | V48 to V50, D1 | txid conversation, Imgur, proof_of_payment.png downloaded | Claimed payment evidence received |
| S8 Blockchain lookup | 14:58:59 to 14:59:21 | V51 to V58 | Explorer search of same txid, then an address | Payment verification attempt |
| S9 Exchange access | 14:59:38 to 15:02:05 | V59 to V71 | Kraken sign-in, device approval, Funding, link to same txid on mempool.space, Ledger | Account checked |
| S10 Back to email | 15:02:32 to 15:02:56 | V72 to V74 | txid conversation, two compose URLs | Reply drafted or opened |

Elapsed times worth noting: 4.99 s from posting confirmation to the manage page; 3.3 s to the public listing; 7.5 s to the Gmail search; 2 min 18.3 s from opening the txid conversation to the transaction lookup; and 4 min 33.7 s from opening it to the Kraken Funding page. The whole session spans 1 h 8 min 38.2 s.

## 5.7 Conclusion, confidence and limitations (Question 8)

**Conclusion.** The Chrome history shows one browser profile, on a Windows account named FSCS_User, publishing a "cheaper than Rx supplements" ad on Craigslist Baltimore, signing in to unsub.fscs@gmail.com, handling an order conversation, receiving a link to an image saved as `proof_of_payment.png` with a transaction ID in the email subject, and then checking that transaction ID on a blockchain explorer and through a Kraken account before opening a reply. This pattern is consistent with the suspected scenario of an online sale paid for in cryptocurrency. It is not sufficient on its own to establish that a controlled substance was sold, what the money was for, or who did it.

| Statement | Confidence | Basis |
|---|---|---|
| The database is intact and the timeline is correct in UTC | High | Matching hashes before and after analysis, integrity_check ok, two independent timestamp conversions, no orphan from_visit |
| This profile created and published Craigslist posting 7473121658 | High | Login, seven FORM_SUBMIT steps, confirmation, manage page and public listing share the posting ID and session key |

| Statement | Confidence | Basis |
|---|---|---|
| This profile signed in to unsub.fscs@gmail.com | High | Identifier, password-challenge and CheckCookie sequence, then inbox titles naming the account |
| The conversations concern an order for the advertised product and a claimed payment | Moderate | Subjects reuse the ad title, plus "order" and "txid"; contents not available |
| The transaction ID checked on the explorer and via Kraken is the one in the email subject | High (match) / Moderate (meaning) | Exact 64-character match in V48, V55 to V57, V67 to V70 and V74; what the transaction was is unverified |
| The ad was for illegal drugs | Low | Only a title and category are recorded |
| A named individual carried out the activity | Low | Evidence identifies a profile, a Windows account and email accounts only |

## Limitations

| # | Limitation | Effect on the conclusion |
|---|---|---|
| 1 | Browser profile is not a person. Nothing in the database shows who was at the keyboard; FSCS_User and the Gmail account could be shared, borrowed or compromised. | Attribution to an individual needs other evidence (account records, device logs, witness). |
| 2 | History stores navigation metadata, not content. No ad text, email bodies, images, amounts or message-sent status. | The nature of the goods, the payment amount and whether a reply was sent cannot be shown. |
| 3 | Titles are stored once per URL and can show a later page title (e.g. Gmail V25 to V29, and the txid thread vs its compose URLs V72, V73). | Titles are used only as supporting labels; the URL, ID and timestamp carry the argument. |
| 4 | Redirects, subframes and single-page-app URL changes create visits the user did not make separately, and History has no tab IDs. | Counts of visits are not counts of user actions; tab usage is inferred only. |
| 5 | History can be cleared or edited, private windows are not recorded, and other browsers or profiles are not covered. The 50 min 48.7 s gap (14:05:59 to 14:56:47) is unexplained. | Absence of a record is not proof that nothing happened. |
| 6 | Timestamps come from the system clock of the machine that wrote them; the time zone is unknown and the clock was not checked against an external source. | Relative order and intervals are reliable; the absolute time is as accurate as that clock. |
| 7 | The brief forbids visiting live artifacts, so the listing, email, image, transaction and address were not checked. | All meanings of those strings are unverified. |
| 8 | The downloaded file was not supplied and its hash was not stored. | The content of proof_of_payment.png is unknown. |

## Recommended follow-up (with proper legal authority)

- Acquire the full Chrome profile and Windows image for the FSCS_User account: Login Data, Cookies, cache, Session files, the Desktop copy of `proof_of_payment.png`, Jump Lists, UserAssist and NTFS timestamps, to confirm who used the machine and whether the file was opened.
- Request account records from Google (Gmail), Craigslist (posting 7473121658 and account) and Kraken (account holder, device approvals, deposits), and read the email bodies lawfully.
- Have blockchain analysis done on the transaction and address by an authorised team, and corroborate the findings with witness or suspect interviews.

## Appendix A: Reconstructed Timeline

All times are UTC on 2022-04-19. "V" is visit ID, "U" is URL ID, "D" is download ID. Observed fact is what the database records; interpretation and limit state what it may mean and what it cannot show.

| Stage | UTC timestamp | Visit or URL ID | Artifact | Observed fact | Interpretation and limit |
|---|---|---|---|---|---|
| S1 | 13:54:18.232 | V1 / U1 | Google results URL; keyword_search_terms | Omnibox search for "craigslist" (transition 805306373, GENERATED). V2 (13:54:19.219) loads the results. | Session starts from a search engine. Search text only. |
| S1 | 13:54:19.946 | V3 to V5 / U2 to U4 | craigslist.org, geo.craigslist.org, baltimore.craigslist.org | Link from results, then two server redirects, all at the same instant, ending on the Baltimore site. | Site chosen by Craigslist's geo redirect; indicates region served, not certain location. |
| S2 | 13:54:22.412 | V6 to V7 / U5, U6 | accounts.craigslist.org/ login/home, /login?rp=... | Account page redirects to login form; login page on screen 17.5 s. | Browser not signed in at that moment. |
| S2 | 13:54:39.868 | V8 / U5 | FORM_SUBMIT to /login/home | Form submitted; page then open 23.4 s. | Consistent with a login; success and account name not recorded. |
| S3 | 13:55:03.307 | V9 to V10 / U2, U4 | craigslist.org, baltimore.craigslist.org | Return to the Baltimore home page (redirect). | Move to the posting area. |
| S3 | 13:55:05.972 | V11 to V13 / U7 to U9 | post.craigslist.org/c/ bal, /k/sHbOUOi_7BG68qUDI beTbQ/bQTi5?s=type | Posting wizard starts; server redirect assigns a session key (V12). | Start of posting in Baltimore. |
| S3 | 13:55:10.245 | V14 / U10 | ?s=cat (FORM_SUBMIT) | Category step submitted, 5.6 s on page. | Category later appears as /hab/ in the listing URL. |
| S3 | 13:55:15.844 | V15 / U11 | ?s=edit (FORM_SUBMIT) | Create-posting form, on page 177.7 s (foreground 178.0 s). | Time fits writing the ad; content not recorded. |
| S3 | 13:58:13.519 to 13:58:19.558 | V16 to V18 / U12 to U14 | ?s=geoverify, ? s=editimage, ?s=preview | Map, image and preview steps, each submitted within 1.6 to 4.4 s. | Normal wizard steps; image or map content not recorded. |
| S3 | 13:58:22.757 | V19 / U8 | Posting confirmation (FORM_SUBMIT) | Final submit reaches the confirmation URL (the same URL as V12). | Publication step completed. |
| S4 | 13:58:27.747 | V20 / U15 | post.craigslist.org/ manage/7473121658 | Manage-posting page, posting ID 7473121658, 5.6 s. | Posting is tied to the signed-in Craigslist account. |
| S4 | 13:58:31.093 | V21 / U16 | baltimore.craigslist.org/ hab/d/baltimore-cheaper-than-rx-supplements/ 7473121658.html | Opened with V20 as opener; title "cheaper than Rx supplements - health and beauty - by owner -...". | Ad is live with the same ID. Title only; goods not recorded. |
| S5 | 13:58:38.596 | V22 to V23 / U17 | Google results; keyword_search_terms | Omnibox search for "gmail", 7.5 s after V21. | Switch to email. |
| S5 | 13:58:40.385 | V24 to V30 / U18 to U24 | Gmail redirect chain | Seven visits at one instant: mail.google.com, ServiceLogin, help pages, ending on www.google.com/gmail/about/. | Not signed in. Titles on U19, U20 are from a later load. |
| S5 | 13:58:43.290 to 13:59:26.399 | V31 to V36 / U25 to U29 | AccountChooser, signin/v2/identifier, challenge/pwd | Identifier page 13:58:44.178; password challenge 42.2 s later. | Account identifier entered; password step reached. |
| S5 | 13:59:29.366 | V37 to V42 / U30 to U32, U19, U20, U33 | CheckCookie, SetOSID, SetSID, mail inbox | V36 duration 2,967.5 ms ends exactly at V37; inbox titles name unsub.fscs@gmail.com. | Sign-in completed at about 13:59:29. Account identity from titles. |
| S6 | 14:04:12.566 | V43 / U34 | Gmail conversation "cheaper than Rx supplements - order" | Opened from the inbox 4 min 40.6 s after it loaded. | Order conversation exists; subject only. |

| Stage | UTC timestamp | Visit or URL ID | Artifact | Observed fact | Interpretation and limit |
|---|---|---|---|---|---|
| S6 | 14:04:35.250 and 14:04:46.777 | V44, V45 / U35, U36 | Thread URL with ? compose= (two tokens) | Two compose states, 11.5 s apart (client redirects), then thread again at 14:04:52.145 (V46). | Compose or reply opened twice; sending not shown. |
| S6 | 14:05:59.142 | V47 / U33 | Gmail inbox | Last entry before a gap of 50 min 48.7 s. | Gap unexplained (idle, other app or private window). |
| S7 | 14:56:47.857 | V48 / U37 | Gmail conversation titled "... -- txid: 517b2156...afdc5" | A different thread ID; 64-character transaction ID in the title. | Sender or subject carries a claimed transaction ID. Unverified. |
| S7 | 14:56:52.464 | V49 to V50 / U38, U39 | google.com/url? q=imgur.com/ dTgrkP7&source=gmail, imgur.com/dTgrkP7 | Link followed from V48, redirect to Imgur, page open 18.9 s. | Image link was inside the email. |
| S7 | 14:56:58.494 to 14:57:10.261 | D1 / U39 (tab_url) | downloads id 1, downloads_url_chains | i.imgur.com/dTgrkP7.png saved as C:\Users\FSCS_User\ Desktop\proof_of_payment.png, 33,844 bytes, complete; 6.0 s after V50. | File reached the FSCS_User desktop. opened = 0; content unseen; hash empty. |
| S8 | 14:58:59.580 | V51 to V54 / U40, U41 | Google results, blockchain.com/explorer | Search "blockchain explorer" 1 min 49.3 s after the download ended. | Move to verify the transaction. |
| S8 | 14:59:06.190 | V55 to V57 / U42, U43 | blockchain.com/search? search=517b...afdc5, /btc/tx/517b...afdc5 | Search redirects to the tx page; the tx page appears again 0.7 s later (V57). 2 min 18.3 s after V48. | Same ID as in V48. Transaction not examined. |
| S8 | 14:59:21.433 | V58 / U44 | blockchain.com/btc/ address/ 38RcsURWCDmCbYoc msZCnGz1FtD7x477mt | Address page opened from the explorer (from_visit 54). | An address of interest; owner unknown; not visited. |
| S9 | 14:59:38.025 | V59 to V62 / U45, U46 | Google results, kraken.com/en-us/sign-in | Search "kraken", then the sign-in page (twice). | Exchange access begins. |
| S9 | 15:00:21.272 | V63 / U47 | kraken.com/en-us/ device-approval | "Approve new device", on page 104.4 s, 34.1 s after the sign-in page. | Kraken did not recognise the browser. Approval path not recorded. |
| S9 | 15:01:11.671 to 15:01:21.602 | V64 to V66 / U48 to U50 | kraken.com/u/trade, /u/instant, /u/funding | Funding title "41,543.80 USD - Kraken - Funding - Overview". | Signed-in session. USD value is the last title stored for the URL. |
| S9 | 15:01:31.926 | V67 to V70 / U51 to U54 | kraken.com/redirect? url=...mempool.space/ tx/517b...afdc5, mempool.space/tx/517b.. .afdc5 | Four visits in 1.9 s, opened from V66, ending on the same transaction ID. | Link followed from the Funding page; why not recorded. |
| S9 | 15:02:05.697 | V71 / U55 | kraken.com/u/history/ ledger | Title "41,563.70 USD - Kraken - History - Ledger" (19.90 USD above the Funding title). | Ledger page viewed; amounts not otherwise recorded. |
| S10 | 15:02:32.450 to 15:02:56.453 | V72 to V74 / U56, U57, U37 | txid thread with ? compose= (two tokens), then the thread | Parent is V48; two compose URLs 11.7 s apart, thread again 12.3 s later; compose titles say "- order -". | Reply window opened after the checks; sending not shown. |

## Appendix B: Evidence Log

Figure numbers refer to this report. File names refer to the evidence package (`evidence_records/` and `sql/`).

| ID | Source record | Finding | Why it matters | SQL or screenshot reference |
|---|---|---|---|---|
| E01 | Original History file; stat, ls, sha256sum, md5sum, file output | 196,608 bytes; SHA-256 b991b0fa...cde3fd19; MD5 0c594e9d...a67e2c74; SQLite 3.x, 41 pages in header | Fixes what was received and lets anyone verify it | Figure 2; acquisition_date_utc.txt, file_stat.txt, sha256_original.txt, md5_original.txt, file_type.txt |
| E02 | Working copy hash, re-check after queries | Hashes identical before and after; only History in working/ | Shows the analysis did not change the evidence | Figures 3, 4, 13; sha256_working.txt, sha256_working_after_queries.txt, working_dir_listing.txt |
| E03 | sqlite_master, PRAGMA table_info | 16 tables; schemas of urls, visits, downloads, downloads_url_chains | Q2: tables and fields | Figures 5, 6; 01_schema.sql; schema_and_counts.txt |
| E04 | Row counts, integrity_check, page size/count, meta | urls 57, visits 74, downloads 1, chains 1, searches 4; integrity ok; 41 pages; meta version 53 | Database is sound and complete | Figures 7, 8; 04_db_health.sql; table_row_counts.txt, db_health.txt |
| E05 | Bytes after the declared page count | 28,672 trailing bytes, 0 non-zero | Rules out hidden data in the padding | Figure 9; trailing_bytes_nonzero.txt |
| E06 | visit_time conversion | Raw 13294850058231547 = 2022-04-19 13:54:18 UTC by SQL and by date | Q3: timestamps are right | Figures 12, 13; 02_visit_timeline.sql; timestamp_crosscheck.txt |
| E07 | visits JOIN urls (74 rows); urls export | Complete visit timeline with transitions, durations, titles, URLs | Basis of Q4 and Q7 | Figures 10, 16, 17; visit_timeline_full.csv, urls_full.csv; 07_urls.sql |
| E08 | V8, V11 to V19 (U5, U7 to U14, U8) | Login submit and seven-step posting wizard under one session key, ending in confirmation | Q5: ad created in this profile | Figure 16; visit_timeline_full.csv rows 8, 11 to 19 |
| E09 | V20, V21 (U15, U16) | Manage page and public listing for ID 7473121658; title "cheaper than Rx supplements - health and beauty - by owner" | Q5: ad was live; identifies the goods only by title | Figure 16; rows 20, 21; opener_visit 21 = 20 |
| E10 | keyword_search_terms (4 rows) | craigslist, gmail, blockchain explorer, kraken | Q6: how each service was reached | Figure 18; 06_searches.sql; search_terms.txt |
| E11 | V22 to V42 (U17 to U33) | Failed first Gmail redirect, then identifier, password challenge, CheckCookie and inbox for unsub.fscs@gmail.com | Q6: mailbox accessed by this profile | Figures 16, 17; rows 22 to 42 |
| E12 | V43 to V47 (U33 to U36) | Conversation "... - order" opened; two compose URLs; then a 50 min 48.7 s gap | Q6: order correspondence | Figures 16, 17; rows 43 to 47 |
| E13 | V48, V72 to V74 (U37, U56, U57) | Conversation with txid 517b2156...afdc5 in title; two later compose URLs | Q6: claimed payment and reply | Figure 17; rows 48, 72 to 74 |
| E14 | V49, V50 (U38, U39) | Gmail link to imgur.com/dTgrkP7 via google.com/url?source=gmail | Q6: image reached from the email | Figure 17; rows 49, 50; opener_visit 49 = 48 |
| E15 | downloads id 1; downloads_url_chains id 1 | i.imgur.com/dTgrkP7.png saved as FSCS_User Desktop proof_of_payment.png, 33,844 bytes, complete, 14:56:58 to 14:57:10 UTC | Q6: payment-proof file saved locally | Figure 19; 05_downloads.sql; downloads.txt |
| E16 | V51 to V58 (U40 to U44) | Explorer search of the same txid; address 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt opened | Q6: payment verification attempt | Figure 17; rows 51 to 58 |
| E17 | V59 to V63 (U45 to U47) | Kraken sign-in and "Approve new device" | Q6: exchange account accessed | Figure 17; rows 59 to 63 |

| ID | Source record | Finding | Why it matters | SQL or screenshot reference |
|---|---|---|---|---|
| E18 | V64 to V66, V71 (U48 to U50, U55) | Trade, Instant, Funding and Ledger pages; USD values in titles | Q6: signed-in exchange session | Figure 17; rows 64 to 66, 71 |
| E19 | V67 to V70 (U51 to U54) | Kraken redirect to mempool.space/tx/517b2156...afdc5, opened from V66 | Q6: same txid opened from the exchange | Figure 17; rows 67 to 70; opener_visit 67 = 66 |
| E20 | Recursive from_visit query; opener_visit | 21 chains, no orphan links, 28 openers | Q4: navigation structure | Figures 14, 15; 08_chains.sql; navigation_chains.txt |
| E21 | content_annotations (47), context_annotations (45) | related_searches, search_terms, entities; foreground durations and page_end_reason | Supporting only (e.g. V15 foreground 178.0 s) | Figure 20; content_annotations.csv, context_annotations.csv |
| E22 | First and last visit; V47 to V48 | Session spans 1 h 8 min 38.2 s; 50 min 48.7 s gap | Q7/Q8: overall pattern and a limit | visit_timeline_full.csv rows 1, 47, 48, 74 |
| E23 | Domain summary | 13 hosts; mail.google.com 16 visits, google.com 12, post.craigslist.org 10, kraken.com 10 | Q2/Q7: where the activity went | Figure 11; 03_domains.sql; domain_summary.txt |

# Appendix C: Commands and SQL

## Acquisition and verification

```
mkdir -p ~/SBTDF204_CS2/{original,working,screenshots,evidence_records,exports,sql}
cd ~/SBTDF204_CS2
cp -p /mnt/hgfs/kali_documents/History original/History
date -u | tee evidence_records/acquisition_date_utc.txt
stat original/History | tee evidence_records/file_stat.txt
ls -l original/History | tee evidence_records/file_size.txt
sha256sum original/History | tee evidence_records/sha256_original.txt
md5sum original/History | tee evidence_records/md5_original.txt
file original/History | tee evidence_records/file_type.txt
chmod a-w original/History
ls -l original/History | tee evidence_records/file_perms_after_chmod.txt
cp -p original/History working/History
sha256sum working/History | tee evidence_records/sha256_working.txt
diff <(cut -d' ' -f1 evidence_records/sha256_original.txt) <(cut -d' ' -f1
evidence_records/sha256_working.txt) && echo "HASHES MATCH"
```

## Queries (run as sqlite3 -readonly working/History < sql/NN_name.sql)

```
-- 02_visit_timeline.sql
SELECT v.id AS visit_id, u.id AS url_id,
       strftime('%Y-%m-%d %H:%M:%f', (v.visit_time/1000000.0) - 11644473600, 'unixepoch')
AS visit_utc,
       v.from_visit, v.transition, (v.transition & 255) AS core_type,
       CASE (v.transition & 255) WHEN 0 THEN 'LINK' WHEN 1 THEN 'TYPED' WHEN 2 THEN
'AUTO_BOOKMARK'
         WHEN 3 THEN 'AUTO_SUBFRAME' WHEN 4 THEN 'MANUAL_SUBFRAME' WHEN 5 THEN
'GENERATED(omnibox)'
         WHEN 6 THEN 'START_PAGE' WHEN 7 THEN 'FORM_SUBMIT' WHEN 8 THEN 'RELOAD'
         WHEN 9 THEN 'KEYWORD' WHEN 10 THEN 'KEYWORD_GENERATED' ELSE 'OTHER' END AS
core_name,
       printf('0x%08X', v.transition & 4294967040) AS qualifiers,
       (v.transition & 268435456 <> 0) AS chain_start, (v.transition & 536870912 <> 0) AS
chain_end,
       (v.transition & 1073741824 <> 0) AS client_redirect, (v.transition & 2147483648 <>
0) AS server_redirect,
       v.visit_duration/1000.0 AS duration_ms, u.title, u.url
FROM visits AS v JOIN urls AS u ON v.url = u.id ORDER BY v.visit_time;

-- 05_downloads.sql
SELECT d.id AS download_id, d.guid, d.target_path, d.current_path,
       strftime('%Y-%m-%d %H:%M:%f', (d.start_time/1000000.0) - 11644473600, 'unixepoch')
AS start_utc,
       strftime('%Y-%m-%d %H:%M:%f', (d.end_time/1000000.0) - 11644473600, 'unixepoch') AS
end_utc,
       strftime('%Y-%m-%d %H:%M:%f', (d.last_access_time/1000000.0) - 11644473600,
'unixepoch') AS last_access_utc,
       d.received_bytes, d.total_bytes, d.state, d.danger_type, d.interrupt_reason,
d.opened,
       d.mime_type, d.original_mime_type, d.referrer, d.site_url, d.tab_url,
d.tab_referrer_url, hex(d.hash) AS hash_hex
FROM downloads AS d ORDER BY d.start_time;
```

```
SELECT id, chain_index, url FROM downloads_url_chains ORDER BY id, chain_index;

-- 08_chains.sql
WITH RECURSIVE chain(id, root, depth) AS (
  SELECT id, id, 0 FROM visits WHERE from_visit = 0
  UNION ALL
  SELECT v.id, c.root, c.depth + 1 FROM visits AS v JOIN chain AS c ON v.from_visit =
c.id )
SELECT root AS chain_root, COUNT(*) AS n_visits, group_concat(id, ' > ') AS visit_path
FROM (SELECT * FROM chain ORDER BY root, depth, id) GROUP BY root ORDER BY root;
SELECT 'orphan from_visit' AS check_name, id FROM visits WHERE from_visit <> 0 AND
from_visit NOT IN (SELECT id FROM visits);
SELECT id AS visit_id, opener_visit FROM visits WHERE opener_visit <> 0;

-- 06_searches.sql
SELECT k.url_id, k.term, u.url FROM keyword_search_terms AS k JOIN urls AS u ON k.url_id =
u.id ORDER BY k.url_id;
```

The remaining query files (01_schema.sql, 03_domains.sql, 04_db_health.sql, 07_urls.sql) are in the evidence package.

### Independent timestamp check and padding check

```
RAW=$(sqlite3 -readonly working/History "SELECT visit_time FROM visits WHERE id = 1;")
echo "raw visit_time = $RAW" | tee evidence_records/timestamp_crosscheck.txt
date -u -d "@$((RAW/1000000 - 11644473600))" '+%Y-%m-%d %H:%M:%S UTC' | tee -a
evidence_records/timestamp_crosscheck.txt
tail -c 28672 working/History | tr -d '\000' | wc -c | tee
evidence_records/trailing_bytes_nonzero.txt
```

## Appendix D: Evidence Package and Handling Notes

- The evidence ZIP `SBTDF204_CaseStudy2_Evidence_2025-FWSD-11468.zip` holds this report (DOCX and PDF), the cropped screenshots used as figures, the saved command output in `evidence_records/`, the SQL files in `sql/`, and a README.md. It does **not** contain the original History database, as the brief requires; its SHA-256 is recorded above so a lecturer can verify any copy they hold.
- Screenshots were cropped to the terminal window only. No unrelated personal data was visible in the terminal output; the Windows taskbar, the VMware library pane and unrelated desktop files were removed. The email address, Windows user name, transaction ID and address shown are part of the case record.
- No live service was contacted: no posting, email account, exchange, wallet, address or transaction ID was opened, and no person was contacted.

# Appendix E: Full Visit Listing (74 visits)

UTC times on 2022-04-19. "From" is `from_visit`. Titles and URLs are shortened here; the full values are in `visit_timeline_full.csv` in the evidence package. Download 1 (14:56:58.494 UTC, URL 39 tab) is listed in Section 5.5.

| V | U | UTC | From | Transition (raw, core type) | Qualifiers | Title (as stored) | URL (shortened) |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 13:54:18.232 | 0 | 805306373 GENERATED | 0x30000000 | craigslist - Google Search | https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989U… |
| 2 | 1 | 13:54:19.219 | 0 | 805306368 LINK | 0x30000000 | craigslist - Google Search | https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989U… |
| 3 | 2 | 13:54:19.946 | 2 | 268435456 LINK | 0x10000000 | craigslist: baltimore jobs, apartments, for s… | https://www.craigslist.org/ |
| 4 | 3 | 13:54:19.946 | 3 | -2147483648 LINK | 0x80000000 | craigslist: baltimore jobs, apartments, for s… | https://geo.craigslist.org/ |
| 5 | 4 | 13:54:19.946 | 4 | -1610612736 LINK | 0xA0000000 | craigslist: baltimore jobs, apartments, for s… | https://baltimore.craigslist.org/ |
| 6 | 5 | 13:54:22.412 | 5 | 268435456 LINK | 0x10000000 | craigslist account | https://accounts.craigslist.org/login/home |
| 7 | 6 | 13:54:22.412 | 6 | -1610612736 LINK | 0xA0000000 | craigslist - account log in | https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L |
| 8 | 5 | 13:54:39.868 | 7 | 805306375 FORM_SUBMIT | 0x30000000 | craigslist account | https://accounts.craigslist.org/login/home |
| 9 | 2 | 13:55:03.307 | 8 | 268435456 LINK | 0x10000000 | craigslist: baltimore jobs, apartments, for s… | https://www.craigslist.org/ |
| 10 | 4 | 13:55:03.307 | 9 | -1610612736 LINK | 0xA0000000 | craigslist: baltimore jobs, apartments, for s… | https://baltimore.craigslist.org/ |
| 11 | 7 | 13:55:05.972 | 10 | 268435456 LINK | 0x10000000 | baltimore \| choose type | https://post.craigslist.org/c/bal |
| 12 | 8 | 13:55:05.972 | 11 | -2147483648 LINK | 0x80000000 | baltimore \| posting confirmation | https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5 |
| 13 | 9 | 13:55:05.972 | 12 | -1610612736 LINK | 0xA0000000 | baltimore \| choose type | https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=ty… |
| 14 | 10 | 13:55:10.245 | 13 | 805306375 FORM_SUBMIT | 0x30000000 | baltimore \| choose category | https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=cat |
| 15 | 11 | 13:55:15.844 | 14 | 805306375 FORM_SUBMIT | 0x30000000 | baltimore \| create posting | https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=ed… |
| 16 | 12 | 13:58:13.519 | 15 | 805306375 FORM_SUBMIT | 0x30000000 | baltimore \| add map | https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=ge… |
| 17 | 13 | 13:58:17.908 | 16 | 805306375 FORM_SUBMIT | 0x30000000 | baltimore \| create posting | https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=ed… |
| 18 | 14 | 13:58:19.558 | 17 | 805306375 FORM_SUBMIT | 0x30000000 | baltimore \| create posting | https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=pr… |
| 19 | 8 | 13:58:22.757 | 18 | 805306375 FORM_SUBMIT | 0x30000000 | baltimore \| posting confirmation | https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5 |
| 20 | 15 | 13:58:27.747 | 19 | 805306368 LINK | 0x30000000 | baltimore \| manage posting | https://post.craigslist.org/manage/7473121658 |
| 21 | 16 | 13:58:31.093 | 0 | 805306368 LINK | 0x30000000 | cheaper than Rx supplements - health and beau… | https://baltimore.craigslist.org/hab/d/baltimore-cheaper-than-r… |
| 22 | 17 | 13:58:38.596 | 0 | 838860805 GENERATED | 0x32000000 | gmail - Google Search | https://www.google.com/search?q=gmail&rlz=1C1ONGR_enUS989US989&… |
| 23 | 17 | 13:58:39.542 | 0 | 805306368 LINK | 0x30000000 | gmail - Google Search | https://www.google.com/search?q=gmail&rlz=1C1ONGR_enUS989US989&… |
| 24 | 18 | 13:58:40.385 | 23 | 268435456 LINK | 0x10000000 | Gmail: Free, Private & Secure Email \| Google … | https://www.google.com/gmail/ |
| 25 | 19 | 13:58:40.385 | 24 | -2147483648 LINK | 0x80000000 | Inbox (12) - unsub.fscs@gmail.com - Gmail | https://mail.google.com/mail/ |
| 26 | 20 | 13:58:40.385 | 25 | -2147483648 LINK | 0x80000000 | Inbox (12) - unsub.fscs@gmail.com - Gmail | https://mail.google.com/mail/u/0/ |
| 27 | 21 | 13:58:40.385 | 26 | -2147483648 LINK | 0x80000000 | Gmail: Free, Private & Secure Email \| Google … | https://accounts.google.com/ServiceLogin?service=mail&passive=1… |
| 28 | 22 | 13:58:40.385 | 27 | -2147483648 LINK | 0x80000000 | Gmail: Free, Private & Secure Email \| Google … | https://mail.google.com/intl/en/mail/help/about.html |

| V | U | UTC | From | Transition (raw, core type) | Qualifiers | Title (as stored) | URL (shortened) |
|---|---|---|---|---|---|---|---|
| 29 | 23 | 13:58:40.385 | 28 | -2147483648 LINK | 0x80000000 | Gmail: Free, Private & Secure Email \| Google … | https://www.google.com/intl/en/mail/help/about.html |
| 30 | 24 | 13:58:40.385 | 29 | -1610612736 LINK | 0xA0000000 | Gmail: Free, Private & Secure Email \| Google … | https://www.google.com/gmail/about/ |
| 31 | 25 | 13:58:43.290 | 0 | 268435456 LINK | 0x10000000 | Gmail | https://accounts.google.com/AccountChooser/signinchooser?servic… |
| 32 | 26 | 13:58:43.290 | 31 | -2147483648 LINK | 0x80000000 | Gmail | https://accounts.google.com/AccountChooser?service=mail&continu… |
| 33 | 27 | 13:58:43.290 | 32 | -1610612736 LINK | 0xA0000000 | Gmail | https://accounts.google.com/ServiceLogin?continue=https%3A%2F%2… |
| 34 | 27 | 13:58:44.028 | 0 | 805306368 LINK | 0x30000000 | Gmail | https://accounts.google.com/ServiceLogin?continue=https%3A%2F%2… |
| 35 | 28 | 13:58:44.178 | 0 | 805306368 LINK | 0x30000000 | Gmail | https://accounts.google.com/signin/v2/identifier?continue=https… |
| 36 | 29 | 13:59:26.399 | 0 | 805306368 LINK | 0x30000000 | Gmail | https://accounts.google.com/signin/v2/challenge/pwd?continue=ht… |
| 37 | 30 | 13:59:29.366 | 36 | 268435456 LINK | 0x10000000 | Inbox (12) - unsub.fscs@gmail.com - Gmail | https://accounts.google.com/CheckCookie?continue=https%3A%2F%2F… |
| 38 | 31 | 13:59:29.366 | 37 | -2147483648 LINK | 0x80000000 | Inbox (12) - unsub.fscs@gmail.com - Gmail | https://mail.google.com/accounts/SetOSID?authuser=0&continue=ht… |
| 39 | 32 | 13:59:29.366 | 38 | -2147483648 LINK | 0x80000000 | Inbox (12) - unsub.fscs@gmail.com - Gmail | https://accounts.youtube.com/accounts/SetSID?ssdc=1&sidt=ALWU2c… |
| 40 | 19 | 13:59:29.366 | 39 | -2147483648 LINK | 0x80000000 | Inbox (12) - unsub.fscs@gmail.com - Gmail | https://mail.google.com/mail/ |
| 41 | 20 | 13:59:29.366 | 40 | -2147483648 LINK | 0x80000000 | Inbox (12) - unsub.fscs@gmail.com - Gmail | https://mail.google.com/mail/u/0/ |
| 42 | 33 | 13:59:31.970 | 41 | 1610612736 LINK | 0x60000000 | Inbox (12) - unsub.fscs@gmail.com - Gmail | https://mail.google.com/mail/u/0/#inbox |
| 43 | 34 | 14:04:12.566 | 0 | 268435456 LINK | 0x10000000 | cheaper than Rx supplements - order - unsub.f… | https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGx… |
| 44 | 35 | 14:04:35.250 | 43 | 1073741824 LINK | 0x40000000 | cheaper than Rx supplements - order - unsub.f… | https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGx… |
| 45 | 36 | 14:04:46.777 | 44 | 1073741824 LINK | 0x40000000 | cheaper than Rx supplements - order - unsub.f… | https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGx… |
| 46 | 34 | 14:04:52.145 | 45 | 1610612736 LINK | 0x60000000 | cheaper than Rx supplements - order - unsub.f… | https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGx… |
| 47 | 33 | 14:05:59.142 | 0 | 805306368 LINK | 0x30000000 | Inbox (12) - unsub.fscs@gmail.com - Gmail | https://mail.google.com/mail/u/0/#inbox |
| 48 | 37 | 14:56:47.857 | 0 | 268435456 LINK | 0x10000000 | cheaper than Rx supplements -- txid: 517b2156… | https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtRjhGdkMB… |
| 49 | 38 | 14:56:52.464 | 0 | 268435456 LINK | 0x10000000 | Imgur: The magic of the Internet | https://www.google.com/url?q=https://imgur.com/dTgrkP7&source=g… |
| 50 | 39 | 14:56:52.464 | 49 | -1610612736 LINK | 0xA0000000 | Imgur: The magic of the Internet | https://imgur.com/dTgrkP7 |
| 51 | 40 | 14:58:59.580 | 0 | 838860805 GENERATED | 0x32000000 | blockchain explorer - Google Search | https://www.google.com/search?q=blockchain+explorer&rlz=1C1ONGR… |
| 52 | 40 | 14:59:00.633 | 0 | 805306368 LINK | 0x30000000 | blockchain explorer - Google Search | https://www.google.com/search?q=blockchain+explorer&rlz=1C1ONGR… |
| 53 | 41 | 14:59:01.659 | 52 | 805306368 LINK | 0x30000000 | Blockchain Explorer - Search the Blockchain \|… | https://www.blockchain.com/explorer |
| 54 | 41 | 14:59:02.754 | 0 | 805306368 LINK | 0x30000000 | Blockchain Explorer - Search the Blockchain \|… | https://www.blockchain.com/explorer |
| 55 | 42 | 14:59:06.190 | 54 | 268435456 LINK | 0x10000000 | Transaction: 517b2156914944339a96137ad8978408… | https://www.blockchain.com/search?search=517b2156914944339a9613… |
| 56 | 43 | 14:59:06.190 | 55 | -1610612736 LINK | 0xA0000000 | Transaction: 517b2156914944339a96137ad8978408… | https://www.blockchain.com/btc/tx/517b2156914944339a96137ad8978… |
| 57 | 43 | 14:59:06.906 | 54 | 805306368 LINK | 0x30000000 | Transaction: 517b2156914944339a96137ad8978408… | https://www.blockchain.com/btc/tx/517b2156914944339a96137ad8978… |
| 58 | 44 | 14:59:21.433 | 54 | 805306368 LINK | 0x30000000 | Address: 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt \|… | https://www.blockchain.com/btc/address/38RcsURWCDmCbYocmsZCnGz1… |
| 59 | 45 | 14:59:38.025 | 0 | 838860805 GENERATED | 0x32000000 | kraken - Google Search | https://www.google.com/search?q=kraken&rlz=1C1ONGR_enUS989US989… |
| 60 | 45 | 14:59:39.513 | 0 | 805306368 LINK | 0x30000000 | kraken - Google Search | https://www.google.com/search?q=kraken&rlz=1C1ONGR_enUS989US989… |

| V | U | UTC | From | Transition (raw, core type) | Qualifiers | Title (as stored) | URL (shortened) |
|---|---|---|---|---|---|---|---|
| 61 | 46 | 14:59:47.217 | 60 | 805306368 LINK | 0x30000000 | Sign in to Kraken - Kraken \| Buy, Sell and Ma… | https://www.kraken.com/en-us/sign-in |
| 62 | 46 | 14:59:47.602 | 0 | 805306368 LINK | 0x30000000 | Sign in to Kraken - Kraken \| Buy, Sell and Ma… | https://www.kraken.com/en-us/sign-in |
| 63 | 47 | 15:00:21.272 | 0 | 805306368 LINK | 0x30000000 | Approve new device - Kraken \| Buy, Sell and M… | https://www.kraken.com/en-us/device-approval |
| 64 | 48 | 15:01:11.671 | 63 | 805306368 LINK | 0x30000000 |  | https://www.kraken.com/u/trade?i=true |
| 65 | 49 | 15:01:12.688 | 63 | 805306368 LINK | 0x30000000 | Kraken \| Buy, Sell and Margin Trade Bitcoin (… | https://www.kraken.com/u/instant |
| 66 | 50 | 15:01:21.602 | 63 | 805306368 LINK | 0x30000000 | 41,543.80 USD - Kraken - Funding - Overview | https://www.kraken.com/u/funding |
| 67 | 51 | 15:01:31.926 | 0 | 268435456 LINK | 0x10000000 | Transaction: 517b2156914944339a96137ad8978408… | https://www.kraken.com/redirect?url=https%3A%2F%2Fmempool.space… |
| 68 | 52 | 15:01:33.808 | 67 | 1073741824 LINK | 0x40000000 | Transaction: 517b2156914944339a96137ad8978408… | http://www.kraken.com/redirect?t=-_F22IhwssFD |
| 69 | 53 | 15:01:33.808 | 68 | -2147483648 LINK | 0x80000000 | Transaction: 517b2156914944339a96137ad8978408… | https://www.kraken.com/redirect?t=-_F22IhwssFD |
| 70 | 54 | 15:01:33.808 | 69 | -1610612736 LINK | 0xA0000000 | Transaction: 517b2156914944339a96137ad8978408… | https://mempool.space/tx/517b2156914944339a96137ad8978408ea52b2… |
| 71 | 55 | 15:02:05.697 | 63 | 805306368 LINK | 0x30000000 | 41,563.70 USD - Kraken - History - Ledger | https://www.kraken.com/u/history/ledger |
| 72 | 56 | 15:02:32.450 | 48 | 1073741824 LINK | 0x40000000 | cheaper than Rx supplements - order - unsub.f… | https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtRjhGdkMB… |
| 73 | 57 | 15:02:44.171 | 72 | 1073741824 LINK | 0x40000000 | cheaper than Rx supplements - order - unsub.f… | https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtRjhGdkMB… |
| 74 | 37 | 15:02:56.453 | 73 | 1610612736 LINK | 0x60000000 | cheaper than Rx supplements -- txid: 517b2156… | https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtRjhGdkMB… |